Notes on Gravity

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Preface

General Relativity is one of the most successful theories of Physics in history. It seems to hold up to every experimental test thrown at it, even more than a century after its conception. In undergraduate lectures this topic is often presented alongside an introduction to tensor calculus and differential geometry. In contrast, these notes will not contain this background information. The aim is to focus on the physical consequences of the theory rather than provide a pedagogical overview, and will therefore assume knowledge of some Riemannian geometry. The main reference for these notes is the incredibly comprehensive book by Straumann [Str13], though only a small fraction of the content will be covered here. For chapter 4 the aim is to understand the nature of Lorentzian geometry in contrast to the Riemannian case. Witten has some very approachable notes [Wit20] as does Galloway [Gal14]. The focus will be on the causal structure of spacetime and singularity theorems. These notes are a work in progress, so there will be comments spotted throughout for areas which need to be filled in. Time permitting, I aim to keep this as a longer term project so that I can discuss more exotic features of gravity.

Contents

1 Setup of General Relativity **3**
- 1.1 Motivation . . . 3
- 1.2 Einstein-Hilbert Action . . . 4
- 1.3 Geodesic Flow . . . 6
- 1.4 Spacetime . . . 10
  - 1.4.1 The Hole argument . . . 10
  - 1.4.2 General Covariance and Diffeomorphism Invariance . . . 10
- 1.5 Moving Frames & Vielbeins . . . 10

2 Schwarzschild Spacetime **12**
- 2.1 Derivation of the Metric . . . 12
- 2.2 Chart Problems . . . 15
- 2.3 Geodesics in Schwarzschild Spacetime . . . 16

3 Kerr Spacetime **18**
- 3.1 Metric . . . 18
- 3.2 Analysing the Geometry of Kerr Spacetime . . . 18
- 3.3 Integrability of Kerr Geodesic Flow . . . 18
  - 3.3.1 Hamilton-Jacobi equation . . . 19

4 Causality & Topology **20**
- 4.1 Preliminaries . . . 20
- 4.2 Causality Conditions . . . 22
- 4.3 Hawking's Theorem . . . 22
- 4.4 Penrose's Theorem . . . 22
- 4.5 String Theory and Instability . . . 22

Chapter 1

Setup of General Relativity

1.1 Motivation

This section will provide an overview of the framework of Einstein's theory. The starting point is to understand the theory of special relativity. We assume that the three dimensions of space and one dimension of time can be treated as a combined space, $\mathbb{R}^4$, equipped with an inner product or metric η. The key distinction between ordinary Euclidean inner product spaces is that the metric is no longer positive definite. The metric can be put in diagonal form and the number of $\pm$ eigenvalues determine its signature. We say that a Lorentzian metric has signature (p, q) if there are p positive eigenvalues and q negative eigenvalues. In the context of special relativity we demand that the metric has signature $(1, 3)$ or $(3, 1)$. Both of the above choices are equivalent, but we will choose to work with the second, so the metric has diagonal form $(-1, +1, +1, +1)$. We write $\mathbb{R}^{1,3}$ when referring to four dimensional Cartesian space equipped with a metric of signature $(3, 1)$ or $(1, 3)$. Roughly speaking the reason for this is that the indefinite signature allows us to distinguish a 'time direction' in spacetime based on the signature. For example a vector $v \in \mathbb{R}^{1,3}$ is 'pointing in the time direction' if $\eta(v, v) < 0$ in our choice of signature. Actually the vector could be pointing 'up' in the time axis or 'down', we need to prescribe which direction time 'flows' in and this is given by a time orientation. In this case it is just choice a constant vector, T, with $\eta(T, T) < 0$.

Definition 1. ***Minkowski spacetime*** *is* $\mathbb{R}^{1,3}$ *equipped with the* ***Minkowski metric*** $\eta(v, v) = -c^2 v_0^2 + \sum_{i=1}^{3} v_i^2$ *and a time orientation.*

With any metric we can associate the orthogonal group $O(p, q)$, in this case $O(3, 1)$, which is the group linear transformations on $\mathbb{R}^{1,3}$ which preserve η. This is the **Lorentz group**. Not all of these transformations are considered to be physical. We restrict to the **proper orthochronous** Lorentz group $SO^+(3, 1)$ which only includes those transformations which have positive determinant and do not allow time and space directions to be reflected. In special relativity observers moving at constant velocity define a so-called inertial reference frame. Any other inertial frame is related to by a transformation in the above group. The transformations can also act vectors with $\eta(v, v) = 0$. In two dimensions $\mathbb{R}^{1,1}$ with time on the y-axis and space on the x, this relation defines two lines crossing the origin with gradient $\pm 1/c$; in other words light rays with velocity c. In this way we recover one of the postulates of special relativity that the speed of light must be constant in all frames of reference.

Furthermore, another postulate of special relativity is that that physical laws should not depend on the choice of inertial reference frame. This can be enforced by demanding that physical quantities should be described by representations of the appropriate Lorentz group. For example, vectors in Minkowski space which transform in the Lorentz group are **four-vectors**. This definition can be extended to apply to tensor quantities too. To get to General Relativity (GR) we suppose that spacetime is no longer described by a Cartesian space but has the property of being curved. To enforce the equivalence principle we demand that no matter how curved spacetime is, it must locally look flat.

This leads to considering that spacetime must be described by a manifold, M, instead of a vector space. It also ought to be a smooth manifold to permit the methods of calculus to be used. Moreover, we want that special relativity should hold locally at each point in space, so it should 'look like' Minkowski spacetime at any point. From Riemannian geometry we know that a manifold at a point looks like its tangent space at that point, the content of the normal neighbourhood theorem. It makes sense to then demand that our tangent spaces are actually Minkowski spaces. This idea is part of Cartan geometry —[REF]—, but cutting to the chase we want a metric tensor defined pointwise as $g_p : T_pM \times T_pM \to \mathbb{R}$ which has a $(3,1)$- signature. Then we define a frame bundle —-[clarify all this]—. The upshot is that tensor fields ought to sections of the associated bundle, to ensure that they transform correctly.

Definition 2 (Spacetime)**.** *Spacetime is tuple* $(M, g, \bigtriangledown, T, \mathrm{d}\mu_g)$.

1. *M is smooth four dimensional manifold;*
2. *$g \in \mathrm{Sym}^2(T^\star M)$ is a Lorentzian metric with signature $(-,+,+,+)$;*
3. *$\bigtriangledown$ is the Levi-Civita connection associated to g;*
4. *T is a time orientation vector field;*
5. *and $\mathrm{d}\mu_g$ is a volume form on M.*

Remark 1. *A smooth manifold may not admit a Lorentzian metric, despite always admitting a Riemannian one.*

Remark 2. *An orientable manifold may not admit a time orientation.*

Because of the above two facts, the spacetimes which are allowed is restricted somewhat. Note that in our definition of spacetime we know require a time orientation **vector field** as opposed to a constant vector in Minkowski spacetime. This makes sense as spacetime is now 'changing' point to point, and hence the tangent spaces at each point are changing. More about causality and the Lorentzian geometry is discussed in chapter 4.

1.2 Einstein-Hilbert Action

General relativity is a classical field theory, where the dynamical field is the metric tensor. This is a crucial difference from other forces modelled by quantum gauge theories, which take place on a fixed background. Einstein originally deduced the equations of motion from physical principles. Another route which was taken by Hilbert —-[and maybe also Einstein]— is to guess an appropriate Lagrangian and derive the equations of motion from the principle of least action.

Definition 3. *The Einstein-Hilbert Action on a Spacetime M is :*

$$S[g] = \int_M \star_g \left(\frac{1}{2\kappa} R + \mathcal{L}_{\mathcal{M}} \right). \tag{1.1}$$

1. $\kappa = 8\pi G c^{-4}$. *Where G is the gravitational constant and c is the speed of light;*
2. $\star_g : \Omega^k(M) \to \Omega^{n-k}(M)$ *is the Hodge dual for the metric g;*
3. *R is the Ricci scalar;*
4. $\mathcal{L}_M$ *is a matter Lagrangian describing the fields and particle distribution.*

It's natural to ask why this specific Lagrangian and not another. One could just say that it's the one that gets the correct equations of motion, but there are multiple ways to justify it from physical principles. —[REF]—. From here on we will make use of local coordinates on spacetime when needed. Denote the chart around a neighbourhood $p \in U \subseteq M$ by $x^\mu \in \mathbb{R}^4$ where $\mu = \{0, 1, 2, 3\}$. Correspondingly, the tangent space basis vectors will be denoted $\{\partial_\mu\}$ and the cotangent basis $\{\mathrm{d}x^\mu\}$. The Einstein summation convention will always be assumed unless stated otherwise.

Definition 4. *The energy momentum tensor, $T_{\mu\nu}$, is defined in terms of the matter Lagrangian by:*

$$T_{\mu\nu} := \frac{-2}{\sqrt{-g}} \frac{\delta}{\delta g^{\mu\nu}} \left[\sqrt{-g} \mathcal{L}_{\mathcal{M}} \right] \tag{1.2}$$

The energy momentum tensor encapsulates the fact matter and energy act as a source for gravity. This quantity won't be used much in these notes, but it is obviously of extreme importance since there is matter in the universe! Moreover, the energy momentum tensor also includes contributions from 'fields' such as the electromagnetic field or scalar fields as well as matter. In a certain sense everything gravitates; even gravity.

Theorem 1. *The variation of the action $\delta S = 0$ implies Einstein's field equations in local coordinates $x^\mu(p)$:*

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu} \tag{1.3}$$

Proof. The variation of the matter Lagrangian is almost by definition the energy momentum tensor so we'll focus on only the first term of the Lagrangian density.

$$\delta(\star_g R) = \delta\left(\sqrt{g} g^{\mu\nu} R_{\mu\nu}\right) = \delta\left(R_{\mu\nu}\right) \sqrt{g} g^{\mu\nu} + \delta\left(\sqrt{g} g^{\mu\nu}\right) R_{\mu\nu} \tag{1.4}$$

Let's compute the second term first. Consider the variation of the inverse metric $\delta(g^{\mu\nu})$. We start by using the definition of the inverse,

$$g^{\mu\sigma} g_{\sigma\nu} = \delta^\mu_\nu, \tag{1.5}$$

now we can apply the variation to both sides and rearrange:

$$\delta(g^{\mu\nu} g_{\nu\sigma}) = \delta(g^{\mu\nu}) g_{\nu\sigma} + g^{\mu\nu} \delta(g_{\nu\sigma}) = 0 \tag{1.6}$$

$$\delta(g^{\mu\nu}) g_{\nu\sigma} = -g^{\mu\nu} \delta(g_{\nu\sigma}) \tag{1.7}$$

$$\delta(g^{\mu\nu}) g_{\nu\sigma} g^{\sigma\rho} = -g^{\mu\nu} \delta(g_{\nu\sigma}) g^{\sigma\rho} \tag{1.8}$$

$$\delta(g^{\mu\rho}) = -g^{\mu\nu} \delta(g_{\nu\sigma}) g^{\sigma\rho} \tag{1.9}$$

Next observe we need to work out the variation of the determinant of a metric. This follows from Jacobi's formula,

$$\delta(g) = g\mathrm{Tr}(g^{\mu\alpha}\delta(g_{\alpha\nu})), \tag{1.10}$$

where we can use the chain rule to obtain :

$$\delta\sqrt{-g} = \frac{1}{2\sqrt{-g}}(-g)g^{\mu\nu}\delta(g_{\mu\nu}) \tag{1.11}$$

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g}g^{\mu\nu}\delta(g_{\mu\nu}) \tag{1.12}$$

We can simplify the calculation of $\delta R_{\mu\nu}$ by using Riemann normal coordinates at $p \in U$, meaning the Christoffel symbols vanish, leaving only

$$\delta R_{\mu\nu} = \partial_\alpha\left(\delta\Gamma^\alpha_{\mu\nu}\right) - \partial_\nu\left(\delta\Gamma^\alpha_{\mu\alpha}\right). \tag{1.13}$$

We can now swap $\partial \to \nabla$ since $\delta\Gamma$ are tensor quantities. This result is the Platini identity.

$$\delta R_{\mu\nu} = \nabla_\alpha\delta\Gamma^\alpha_{\mu\nu} - \nabla_\nu\delta\Gamma^\alpha_{\mu\alpha} \tag{1.14}$$

—[COMPLETE, NOTE BOUNDARY TERM]— □

The field equations are incredibly difficult to solve. They are non-linear, 2nd order, coupled partial differential equations in terms of the metric. The only way to get analytic solutions is in highly symmetric toy models, as we will see in the next two chapters. Yet even then it can be cumbersome! One way to simplify the situation is to consider the case of a vanishing energy momentum tensor.

$$\begin{aligned} R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 0 \\ \implies R - \frac{1}{2}R\delta^\mu_\mu &= 0 \\ \implies R &= 0 \end{aligned}$$

Then we can substitute $R = 0$ back into the vacuum field equations to get,

$$R_{\mu\nu} = 0, \tag{1.15}$$

which is the common form Einstein's vacuum field equations. These notes will be focused mostly on this restricted regime.

1.3 Geodesic Flow

We can model the motion of small objects travelling in spacetime in the absence of forces with geodesics. Since the mass is assumed small we can ignore the back reaction of the mass of the matter on the spacetime/gravitational field. Recall that a **symplectic form** is a closed non-degenerate two form. A manifold with equipped with a symplectic form is a **symplectic manifold**. When

such a space also has a designated a smooth energy function, it is a Hamiltonian system. It will be useful to formalise the dynamics here as a Hamiltonian system on the phase space of spacetime, $T^\star M$. We may also make use of the bundle isomorphism $T^\star M \cong TM$ induced by the metric as $(x^\mu, p_\mu) \mapsto (x^\mu, g^{\mu\nu} p_\nu)$. This lets us pull back the symplectic form to the tangent bundle as,

$$\omega_g = \mathrm{d}x^\mu \wedge (g_{\mu\nu} p^\nu)\,,$$

and then one can consider Lagrangian mechanics. We will stick to Hamilton's formulation for these upcoming definitions.

Definition 5 (Geodesic Flow)**.** *We have a Hamiltonian system, $(T^\star M, \omega, H)$, where in local coordinates (x^μ, p_μ) we define :*

1. *The **symplectic form**, $\omega \in \Omega^2(T^\star M)$, on $T^\star M$ in local coordinates is given by $\omega = \mathrm{d}x^\mu \wedge \mathrm{d}p_\mu$.*
2. *The Hamiltonian is $H = \frac{1}{2} g^{\mu\nu} p_\mu p_\nu$*

Definition 6. *$\exists!\ X_H \in \mathfrak{X}(T^\star M)$ called the **Hamiltonian vector field** satisfying,*

$$\iota_{X_H} \omega = \mathrm{d}H, \tag{1.16}$$

defining Hamilton's equations of motion.

So, for geodesic flow, the symplectic form is just the canonical one. We need a conserved energy to define a Hamiltonian system, here it is just the usual energy defined from the square of momentum, except using the induced metric. Note that we will identify $g^{\mu\nu} p_\nu = p^\mu := \dot{x}^\mu$ as the **four-momentum**. We will write trajectories in spacetime as $x^\mu(\tau)$ as a slight abuse of notation for the coordinates of the curve in a chart, $(x^\mu \circ \gamma)(\tau)$. We then write the components of the tangent vector at a point as $\dot{x}^\mu \partial_\mu$.

Theorem 2. *Geodesic flow is satisfied by a trajectory $x^\mu(\tau) \iff$*

$$\nabla_{\dot\gamma} \dot\gamma = 0. \tag{1.17}$$

$$\iff \ddot{x}^\mu + \Gamma^\mu_{\sigma\nu} \dot{x}^\sigma \dot{x}^\nu = 0 \tag{1.18}$$

Proof. Start with the definition of Hamilton's equations of motion 6, which is equivalent to

$$\dot{x}^\mu = \frac{\partial H}{\partial p_\mu}, \quad \dot{p}_\mu = -\frac{\partial H}{\partial x^\mu}.$$

Then we obtain

$$\dot{x}^\mu = g^{\mu\nu} p_\nu, \quad \dot{p}_\mu = -\frac{1}{2} \partial_\sigma (g^{\mu\nu}) p_\mu p_\nu.$$

Now take the derivative of the left-hand equation to get,

$$\begin{aligned} \ddot{x}^\mu &= g^{\mu\nu} \dot{p}_\nu + \dot{x}^\sigma \partial_\sigma g^{\mu\nu} p_\nu \\ &= g^{\mu\nu} \dot{p}_\nu + \dot{x}^\sigma \partial_\sigma g^{\mu\nu} g_{\nu\rho} \dot{x}^\rho, \end{aligned}$$

where we've used chain rule $\frac{\mathrm{d}}{\mathrm{d}\tau} = \dot{x}^\sigma \partial_\sigma$, and the definition of four momentum $p_\nu = g_{\mu\rho} \dot{x}^\rho$. Next substitute the equation for $\dot{p}$ to get,

$$\ddot{x}^\mu = -\frac{1}{2}\partial_\nu g^{\gamma\delta} p_\gamma p_\delta g^{\mu\nu} + \dot{x}^\sigma \partial_\sigma g_{\nu\rho}\dot{x}^\rho.$$

We need to eliminate the derivative of the inverse metric so note from product rule we have :

$$\begin{aligned}
\partial_\gamma(g_{\mu\nu}g^{\nu\sigma}) &= \partial_\gamma(\delta^\sigma_\mu) = 0 \\
&= g_{\mu\nu}\partial_\gamma g^{\nu\sigma} + g^{\nu\sigma}\partial_\gamma g_{\mu\nu} = 0 \\
\implies g^{\mu\rho}g_{\mu\nu}\partial_\gamma g^{\nu\sigma} &= -g^{\mu\rho}g^{\nu\sigma}\partial_\gamma g_{\mu\nu} \\
\implies \partial_\nu g^{\gamma\delta} &= -g^{\mu\gamma}g^{\sigma\delta}\partial_\nu g_{\mu\sigma}.
\end{aligned}$$

Using this identity we obtain,

$$\ddot{x}^\mu = \frac{1}{2}g^{\mu\gamma}g^{\sigma\delta}\partial_\nu g_{\mu\sigma}g_{\beta\gamma}\dot{x}^\beta g_{\alpha\delta}\dot{x}^\alpha g^{\mu\nu} - g^{\alpha\mu}g^{\beta\nu}\partial_\sigma g_{\alpha\beta}g_{\nu\rho}\dot{x}^\sigma\dot{x}^\rho$$

$$\implies \ddot{x}^\mu = \frac{1}{2}g^{\mu\nu}\dot{x}^\sigma\dot{x}^\alpha\partial_\nu g_{\alpha\sigma} - g^{\alpha\mu}\dot{x}^\sigma\dot{x}^\beta\partial_\sigma g_{\alpha\beta}$$

$$\ddot{x}^\mu = \frac{1}{2}g^{\mu\alpha}(\partial_\alpha g_{\sigma\nu} - \partial_\sigma g_{\alpha\nu} - \partial_\nu g_{\sigma\alpha})\dot{x}^\sigma\dot{x}^\nu$$

$$\implies \ddot{x}^\mu + \Gamma^\mu_{\sigma\nu}\dot{x}^\sigma\dot{x}^\nu = 0,$$

where we've relabelled indices and recognised the formula for the Christoffel symbols of a Levi-Civita connection. On the other hand note that by definition of the induced covariant derivative on a curve,

$$\begin{aligned}
\nabla_{\dot{\gamma}}\dot{\gamma} = 0 &\iff \dot{x}^\mu \nabla_{\partial_\mu}\dot{x}^\nu = 0 \\
\iff \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\partial}{\partial x^\mu}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} &+ \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\Gamma^\mu_{\sigma\nu}\frac{\mathrm{d}x^\sigma}{\mathrm{d}\tau} = 0,
\end{aligned}$$

where we've converted to statements for all charts. Then by chain rule we obtain,

$$\ddot{x}^\mu + \Gamma^\mu_{\sigma\nu}\dot{x}^\sigma\dot{x}^\nu = 0.$$

□

Note that we're being somewhat informal in distinguishing between geodesic and so-called co-geodesic flow, since the equations of motion are the same it won't make a difference. For any path of a massive particle in spacetime, geodesic or not, we can define the length by integration of the metric.

Definition 7. *The **length** of curve $\gamma : [a, b] \to M$ in spacetime is defined as :*

$$L(\gamma) := \int_a^b \sqrt{|g(\dot{\gamma}(\lambda), \dot{\gamma}(\lambda))|}\mathrm{d}\lambda$$

In the case of a massive particle we can use this to define the **proper time** elapsed along a trajectory.

Definition 8. *The **proper time** τ for a massive particle is the arc length parameter associated to the length functional.*

Traditionally this is written as $ds^2 = -d\tau^2$. Recall that in Riemannian geometry geodesics are stationary points of the length functional which locally minimise distances (but not necessarily globally). In Lorentzian geometry geodesics are still stationary points of the new length functional, but now we have the interpretation of locally **maximising** elapsed proper time. Intuitively in the case of special relativity, a stationary particle relative to a frame has no time dilation or anything strange. On the other hand wiggling around causes less proper time to elapsed due to time dilation effects. A similar story is true for general relativity, infinitesimally at least. Note that the concept of things being at rest is a lot more subtle in general relativity, since inertial frames can only be defined locally.

Definition 9. *A **conserved quantity**, $f \in C^\infty(T^\star M)$, is constant along geodesic flow,*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(f \circ \gamma)(\tau) = 0, \tag{1.19}$$

where here $\gamma(\tau) = (x^\mu(\tau), p_\mu(\tau))$ is a curve in phase space, with the usual abuse of notation for coordinates.

Definition 10. *A vector field $X \in \mathfrak{X}(T^\star M)$ is a symmetry for the Hamiltonian system if*

1. $X(H) = 0$
2. $\mathcal{L}_X \omega = 0$

Theorem 3 (Noether)**.** *If $X \in \mathfrak{X}(T^\star M)$ is a symmetry and $H^1(T^\star M) = 0$ then $\exists f \in C^\infty(T^\star M)$ such that*

$$\iota_X \omega = \mathrm{d}f, \tag{1.20}$$

where f is a conserved quantity.

Proof. To show such f exists we just expand the Lie derivative using Cartan's magic formula,

$$\mathcal{L}_X \omega = \iota_X \circ \mathrm{d}\omega + \mathrm{d} \circ \iota_X \omega = 0,$$

but we know that $\mathrm{d}\omega = 0$ since ω is symplectic. Then remaining is

$$\mathrm{d} \circ \iota_X \omega = 0$$

which implies, using the triviality of $H^1(T^\star M)$, that $\exists f \in C^\infty(M)$ satisfying $\iota_X \omega = \mathrm{d}f$. Now it remains to show that f is conserved along flows generated by X. We just have to apply the interior product successively to the symplectic form :

$$\iota_X \circ \iota_{X_H} \omega = \iota_X \circ \mathrm{d}H = X(H) = 0$$
$$-\iota_{X_H} \circ \iota_X \omega = -\iota_{X_H} \circ \mathrm{d}f = 0 \implies X_H(f) = 0$$

□

The above is Noether's theorem in Hamiltonian mechanics, which we will make use of this when solving the geodesic equation. —[PERHAPS QUICKLY DEFINE INTEGRABILITY FOR LATER]—Recall that in Riemannian geometry one has the notion of Killing vector fields:

Definition 11. *A **Killing vector field**, $X \in \mathfrak{X}(M)$, satisfies $\mathcal{L}_X g = 0$.*

Equivalently, a Killing vector field can be thought of as generating infinitesimal isometries along its flow. Note that, along a geodesic, we have that a Killing vector field induces a conserved quantity, $g(\dot{\gamma}, X)$. —[PROVE]—.

The above discussion suffices to describe the path a unit mass particle travels through spacetime, in general we also want to consider massless particles such as photons. In the massive case the Hamiltonian defines the normalisation of momenta by $H = -\frac{1}{2}$, in the massless case the hamiltonian is zero since light rays are null $g(\dot{\gamma}, \dot{\gamma}) = 0$. Furthermore, the length function is identiacally zero so we must find a more suitable parameter to describe the distance travelled. We can actually define the paramater as that which satisfies the geodesic equation.

Definition 12. *An **affine parameter**, λ, is a parameter which satisfies the geodesic equation,*

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\sigma\nu} \frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda} = 0.$$

This parameter is unique up to an affine reparameterisation $\lambda' = a\lambda + b$ where $a, b \in \mathbb{R}$. So even along a light ray, there is still a way to track a 'length' along the geodesic.

1.4 Spacetime

In the abstract mathematical sense a manifold is a perfectly well-defined concept. The extent to which spacetime itself coincides with this definition is not immediatey obvious. When Einstein was originally formulating his theory he struggled with this concept. In this section we'll briefly talk about some of these issues and resolutions.

1.4.1 The Hole argument

In general relativity, and physics as whole, we require laws to be **generally covariant**. —[DEFINE]—

1.4.2 General Covariance and Diffeomorphism Invariance

The resolution to the paradox comes from realising that different field configurations have to be deemed equivalent if they are related to each other by an isometry. In other words we must quotient the space of metrics by some —[COMPLETE]—

1.5 Moving Frames & Vielbeins

In Minkowski spacetime there is a group of symmetries which preserves the metric, the isometry group. In particular, we have translational invariance and Lorentz invariance. Combining these two together, formally the semidirect product, gives us the isometry group of Minkowski space; the **Poincaré group** $\mathrm{Iso}(3,1)$. We say this is a global symmetry since it has no dependence on spacetime, i.e. the group elements are constants and not functions. In gauge theory a powerful principle is to take a global symmetry and turn it into a local symmetry, or gauge symmetry. A common example is the theory of electrically charged particles. Gauging the global $U(1)$-symmetry

imposes the condition that an electromagnetic force appears. One can play the same game with the Poincaré symmetry of Minkowski spacetime. Gauging this global symmetry also causes a new force to appear; gravity.

The way to formally describe this process is through the reduction of a structure group. To every smooth manifold one can consider the principal frame bundle over it. Specifically this comes from the fact that every smooth manifold comes equipped with a metric, so one can define an $O(n)$ principal bundle over the space. This $O(n)$-bundle can be thought of as arising from the reduction of the structure group of a principal $GL(n)$-bundle.—[EXPLAIN]—.

The above formalism is extremely insightful, but there's no new physics apparent yet, so what's the point? Beyond a better understanding of the theory, there are at least two good reasons. The first is that it enables us to perform computations much easier. Momentarily we will describe Cartan's method of moving frames, which we will make heavy use of throughout these notes. The second reason is that the universe contains fermions. For reasons we won't describe here, the metric tensor alone isn't enough to describe spinor fields. In order to have a consistent theory, then one must inevitably pass to the vielbein formalism to describe spinors in curved spacetime. This is especially true for supergravity and supersymmetric string theories, in which generalisations of the vielbein play important role.

In practice one works with two fields —[COMPLETE VIELBEIN FORMALISM]—. The coframe σ^α_μ and the spin connection $\omega^\alpha_{\mu\beta}$ and computes derivatives using the structure equations. Differential forms make the computations go a lot smoother than the usual methods due to cancellations from antisymmetry etc. Note that since we have a Lorentzian signature the 'flat' indices are not antisymmetric as in the Riemannian case, but pick up minus signs when swapping certain indices.

Chapter 2

Schwarzschild Spacetime

2.1 Derivation of the Metric

The Schwarzchild solution is a static and spherically symmetric metric aiming to model the spacetime around a massive object. A derivation can be found in [Str13], we will follow this and also include explicit computations for all quantities. By definition, a static spacetime admits a timelike killing vector field, and we can decompose the manifold as a product,

$$M = \mathbb{R}_t \times \Sigma,$$

for some spatial manifold Σ. In particular the metric has the form,

$$g = -f^2 \mathrm{d}t \otimes \mathrm{d}t + h,$$

where $f \in C^\infty(\Sigma)$ and h is a Riemannian metric on Σ. It is a vacuum solution and therefore satisfies $R_{\mu\nu}$. A spacetime is spherically symmetric if there is an $SO(3)$ action which induces an isometry on the metric. Furthermore, we require that the orbits of this action are spacelike hypersuraces. If the spacetime is also static, as in this case, then the action is trivial on $\mathbb{R}_t$. This works when we explicitly make the assumption the timelike vector field is unique, so that there is a distinguished time coordinate. If ∂_t is unique then the $SO(3)$ action must leave it invariant. The orbits of an isomtric $SO(3)$ action are homogenous Riemannian manifolds of two dimensions. In this case it must have constant curvature, which implies that it is geodsically complete. Furthermore, this means is must be simply-connected.

Recall that the standard simply-connected surface is either the 2-sphere, the flat Euclidean plane, or the Hyperbloic plane depending on the sign of the curvature. There is a known result that there exists a covering of our orbits by the standard surface such that the covering map is a local isometry. Since $SO(3)$ is a subgroup of the isometry group of the orbit space, then the curvature must be strictly positive. From this we deduce the orbit space has to be the 2-sphere, S^2. Now we have a foliation of the spatial manifold as $\mathbb{R}_r \times S^2$, where r is the radial coordinate. This **defined** through the area of the orbit sphere at the point $A = 4\pi r^2$. Now we can write an ansatz for the metric.

$$g = -e^{2a(r)} \mathrm{d}t \otimes \mathrm{d}t + e^{2b(r)} \mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 \sin^2(\theta)\, \mathrm{d}\varphi \otimes \mathrm{d}\varphi \tag{2.1}$$

Note that we use exponentials since the functions have to be positive, we demand that $\mathrm{d}t \neq 0 \neq \mathrm{d}r$. Furthermore, we include a factor of two for convenience in computation. The spherical part of the metric is the standard one for a two-sphere of radius r. Now we need to compute the Einstein tensor for the spacetime. We can read the coframes directly from the metric, $\sigma^\alpha = \sigma^\alpha_\mu \mathrm{d}x^\mu$ as:

$$\sigma^0 = e^{a(r)}\mathrm{d}t \quad \sigma^1 = e^{b(r)}\mathrm{d}r \quad \sigma^2 = r\mathrm{d}\theta \quad \sigma^3 = r\sin(\theta)\,\mathrm{d}\phi$$

Taking derivatives we obtain:

$$\mathrm{d}\sigma^0 = a' e^{-b}\sigma^1 \wedge \sigma^0, \quad \mathrm{d}\sigma^1 = 0,$$
$$\mathrm{d}\sigma^2 = \frac{e^{-b}}{r}\sigma^1 \wedge \sigma^2, \quad \mathrm{d}\sigma^3 = \frac{1}{r}\left(e^{-b}\sigma^1 \wedge \sigma^3 + \cot(\theta)\,\sigma^2 \wedge \sigma^3\right)$$

Now we use Cartan's first structure equation, $\mathrm{d}\sigma^\alpha = -\omega^\alpha_\beta \wedge \sigma^\beta$, to determine the connection form.

$$\mathrm{d}\sigma^0 = -\omega^0_1 \wedge \sigma^1 \implies \omega^0_1 = a' e^{-b}\sigma^0 = \omega^1_0$$
$$\mathrm{d}\sigma^2 = -\omega^2_1 \wedge \sigma^1 \implies \omega^2_1 = \frac{e^{-b}}{r}\sigma^2 = -\omega^1_2$$
$$\mathrm{d}\sigma^3 = -\omega^3_1 \wedge \sigma^1 - \omega^3_2 \wedge \sigma^2 \implies \omega^3_1 = \frac{e^{-b}}{r}\sigma^3 = -\omega^1_3,\ \omega^3_2 = \frac{\cot(\theta)}{r}\sigma^3 = -\omega^2_3$$

Then we can compute the curvature 2-form using Cartan's second structure equation, $\Omega^\alpha_\beta = \mathrm{d}\omega^\alpha_\beta + \omega^\alpha_\gamma \wedge \omega^\gamma_\beta$.

$$\begin{aligned}\Omega^0_1 &= \mathrm{d}\omega^0_1 + \omega^0_\gamma \wedge \omega^\gamma_1 = \mathrm{d}\left(a' e^{-b}\right) \wedge \sigma^0 + a' e^{-b}\mathrm{d}\sigma^0 \\ &= \left(a'' e^{-b} - a' b' e^{-b}\right) \wedge \sigma^0 + (a')^2 e^{-b}\sigma^1 \wedge \sigma^0 \\ &\implies \Omega^0_1 = -e^{-2b}\left(a'' - a' b + (a')^2\right)\sigma^0 \wedge \sigma^1\end{aligned}$$

$$\begin{aligned}\Omega^0_2 &= -\omega^0_1 \wedge \omega^1_2 = -a' e^{-b}\sigma^0 \wedge \frac{1}{r}e^{-b}\sigma^2 \\ &\implies \Omega^0_2 = \frac{a'}{r}e^{-2b}\sigma^0 \wedge \sigma^2\end{aligned}$$

$$\begin{aligned}\Omega^0_3 &= \omega^0_\gamma \wedge \omega^\gamma_3 = -a' e^{-b}\sigma^0 \wedge \frac{1}{r}e^{-b}\sigma^3 \\ &\implies \Omega^0_3 = -\frac{a'}{r}e^{-2b}\sigma^0 \wedge \sigma^3\end{aligned}$$

$$\Omega_2^1 = -\mathrm{d}\left(\frac{e^{-b}}{e}\right) \wedge \sigma^2 - \frac{e^{-b}}{r}\mathrm{d}\sigma^2$$
$$= \left(\frac{e^{-b}}{r^2} + \frac{b'}{r}e^{-b}\right)\mathrm{d}r \wedge \sigma^2 - \frac{e^{-2b}}{r}\left(\frac{1}{r} + b'\right)\sigma^1 \wedge \sigma^2 - \frac{e^{-2b}}{r^2}\sigma^1 \wedge \sigma^2$$
$$\implies \Omega_2^1 = \frac{b'}{r}e^{-2b}\sigma^1 \wedge \sigma^2$$

$$\Omega_3^2 = -\mathrm{d}\left(\frac{\cot(\theta)}{r}\right) \wedge \mathrm{d}\sigma^3 - \frac{\cot(\theta)}{r}\mathrm{d}\sigma^3 - \frac{e^{-2b}}{r^2}\sigma^2 \wedge \sigma^3$$
$$= \frac{1}{r^2}\cot(\theta)\,\mathrm{d}r \wedge \sigma^3 + \frac{1}{r\sin^2(\theta)}\mathrm{d}\theta \wedge \sigma^3 - \frac{\cot(\theta)}{r^2}\left(e^{-b}\sigma^1 \wedge \sigma^3 + \cot(\theta)\,\sigma^2 \wedge \sigma^3\right) - \frac{e^{-2b}}{r^2}\sigma^2 \wedge \sigma^3$$
$$\implies \Omega_3^2 = \frac{1 - e^{2b}}{r^2}\sigma^2 \wedge \sigma^3$$

Next we can compute the components of the Ricci tensor using $R_{\alpha\beta} = \Omega_\alpha^\gamma(\pi_\gamma, \pi_\beta)$, where $\{\pi_\alpha\}_{\alpha=0}^3$ is the dual basis to the coframes. For convenience write $A = -e^{-2b}\left(a'' - a'b + (a')^2\right)$.

$$R_{00} = \Omega_0^\gamma(\pi_\gamma, \pi_0) = -A + \frac{2a'}{r}e^{-2b}$$
$$R_{11} = \Omega_1^\gamma(\pi_\gamma, \pi_1) = A + \frac{2b'}{r}e^{-2b}$$
$$R_{22} = \Omega_2^\gamma(\pi_\gamma, \pi_2) = \frac{e^{-2b}}{r}(b' - a') + \frac{1}{r^2} - \frac{e^{-2b}}{r^2} = R_{33}$$

The Ricci scalar is then just the contraction with the Minkowski metric:

$$R = 2A + 4\frac{e^{-2b}}{r}(b' - a') + \frac{2}{r^2} - \frac{2e^{-2b}}{r^2}$$

Finally, the Einstein tensor components, $G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}Rg_{\alpha\beta}$, are as follows :

$$G_{00} = \frac{1}{r^2} - e^{-2b}\left(\frac{1}{r^2} - \frac{2b'}{r}\right)$$
$$G_{11} = -\frac{1}{r^2} + e^{-2b}\left(\frac{1}{r^2} + \frac{2a'}{r}\right)$$
$$G_{22} = e^{-2b}\left((a')^2 - a'b' + a'' + \frac{a' - b'}{r}\right)$$

Now we only have two functions to solve for, namely $a(r)$ and $b(r)$, using $G_{\alpha\beta} = 0$. Note that,

$$G_{00} + G_{11} = \frac{2(b' + a')}{r} = 0 \implies b + a = C, \tag{2.2}$$

but we imposed asymptotic flatness meaning the integration constant C has to be zero to keep $a, b \to 0$. This means we can eliminate $a = -b$ and look at G_{00}.

$$\frac{1}{r^2} - e^{-2b}\left(\frac{1}{r^2} - \frac{2b'}{r}\right) = 0 \tag{2.3}$$

Multiplying through by r^2 we get,

$$1 = e^{-2b} - 2b' r e^{-2b}, \tag{2.4}$$

where we can recognise the right-hand side as $(re^{-2b})'$. Now we can integrate both sides and pick up a constant $2m$,

$$r = re^{-2b} + 2m \implies e^{-2b} = 1 - \frac{2m}{r},$$

leaving us finally with the full metric

$$g = -\left(1 - \frac{2m}{r}\right) \mathrm{d}t \otimes \mathrm{d}t + \left(1 - \frac{2m}{r}\right)^{-1} \mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + r^2 \sin^2(\theta) \mathrm{d}\phi \otimes \mathrm{d}\phi. \tag{2.5}$$

One can show that m should be equal to GM/c^2, where G is Newton's gravitational constant and M has the interpretation as the mass of the central body. The reason is that we expect GR to reduce to Newtonian gravity in a sufficient limiting scenario. This is called the weak field approximation.

2.2 Chart Problems

Importantly, there a two singularities appearing the above metric. The first of which occurs at the so-called event horizon $r = 2m$:

$$r \to 2m \implies g_{tt} \to 0,\ g_{rr} \to \infty$$

The second singularity occurs when $r = 0$:

$$r \to 0 \implies g_{tt} \to \infty,\ g_{rr} \to 0$$

Upon close analysis it's revealed that the first is actually just a coordinate singularity, meaning our choice of chart is at fault here. In particular there exists a change of coordinates such that the metric is well-behaved at the event horizon, which will be discussed momentarily. On the other hand, the singularity at $r = 0$ is very real, no matter what chart we use this is a problematic point in spacetime and must be removed. We must modify our definition of chart to account for this, in particular the image of the r coordinate map should be :

$$\mathrm{Im}(x^1) = (0, 2m) \cup (2m, \infty) \subset \mathbb{R}$$

Now, just because our chart doesn't see the points at $r = 2m$, it doesn't mean that this is not a part of the spacetime. We can extend the chart to cover the whole manifold, excluding the singularity, using the following change of coordinates:

$$u(t,r) = \sqrt{\frac{r}{2m} - 1} e^{r/4m} \cosh\left(\frac{t}{4m}\right) \tag{2.6}$$

$$v(t,r) = \sqrt{\frac{r}{2m} - 1} e^{r/4m} \sinh\left(\frac{t}{4m}\right) \tag{2.7}$$

These are the **Kruskal** transformations. It's not immediately clear where these come from, but they can actually be derived from first principles, as is done in [Str13]. Noting special happens with the θ and φ coordinates so we can ignore those. Analysis of the transformed coordinates shows that u and v are in fact well-defined at $r = 2m$ and there is no longer a singularity. We can now consider the metric in the new coordinates with the points at $r = 2m$ included:

$$g = \frac{32m^3}{r} e^{-r/2m} (\mathrm{d}v \otimes \mathrm{d}v - \mathrm{d}u \otimes \mathrm{d}u) + r^2 (\mathrm{d}\theta \otimes \mathrm{d}\theta + \sin^2(\theta) \mathrm{d}\varphi \otimes \mathrm{d}\varphi) \tag{2.8}$$

The previous discussion highlights that singularities in general relativity are a delicate matter. They will be discussed more in Chapter 4.

2.3 Geodesics in Schwarzschild Spacetime

We will stick to the Lagrangian description for now, which is equivalent to the Hamiltonian description in chapter 1. We will make use of the abuse of notation, $x^\mu(\tau) \coloneqq x^\mu \circ \gamma(\tau)$, for the coordinates along a curve in spacetime, $\gamma : \mathbb{R} \to M$. The Lagrangian is then,

$$2\mathcal{L} = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -1, \tag{2.9}$$

where the dot denotes differentiation with respect to proper time. Beacuse there is spherical symmetry, geodesics are confined to stay on a fixed plane. Without loss of generality we can fix this plane to be $\theta = \pi/2$ to simplify the setup. We then obtain the Lagrangian from the metric as,

$$2\mathcal{L} = -\left(1 - \frac{2m}{r}\right) \dot{t}^2 + \left(1 - \frac{2m}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\varphi}^2. \tag{2.10}$$

Recall that symmetries of the metric, killing vectors, correspond to conserved quantities along geodesic flow. So, before even writing the geodesic equations, we have by definition two conserved quantities:

$$\mathcal{L}_{\partial_t} g = 0 \implies -\left(1 - \frac{2m}{r}\right) \dot{t}^2 \coloneqq -E$$
$$\mathcal{L}_{\partial_\varphi} g = 0 \implies r^2 \dot{\varphi} \coloneqq L$$

These correspond to an energy and angular momentum of the particle. Now we substitute the energy back into eq. (2.10) to obtain,

$$\dot{r}^2 + \left(1 - \frac{2m}{r}\right) \left(1 + \frac{L^2}{r^2} = E^2\right). \tag{2.11}$$

Note also that we can express $\dot{r}$ in terms of $r' := \frac{\mathrm{d}r}{\mathrm{d}\varphi}$ with the chain rule, $\dot{r} = r'\dot{\varphi}$. Then substiuting the formula for angular momentum we have,

$$\frac{\mathrm{d}r}{\mathrm{d}\tau} = \frac{L}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\varphi},$$

which can be substituted back into $2\mathcal{L} = -1$ to get:

$$\left(\frac{\mathrm{d}r}{\mathrm{d}\varphi}\right)^2 \frac{L^2}{r^4} + \left(1 + \frac{2m}{r}\right)\left(1 - \frac{L^2}{m}\right) = E^2. \tag{2.12}$$

This can be solved exactly by using the transformation $u = 1/r$, but we won't do this here.

Chapter 3

Kerr Spacetime

3.1 Metric

Kerr spacetime is again a vacuum solution to the Einstein's field equations. It models the field generated by rotating matter, and like the Schwarzschild case it contains a singularity. In the Schwarzschild case we had a spherical symmetry, in this case we loosen this to an axial one. The metric will also be static, hence there will be something similar to a timelike killing field. Note that this is slightly strange considering the mass is spinning constantly (and hence not 'static'), but we're assuming that the angular momentum is fixed for all time. —[ASYMPTOTIC TIMELIKE]—-

There's not really a quick and conceptually intuitive way to derive this metric, unlike in the Schwarzschild case. There is a derivation in Straumann [Str13] which serves the purpose of being more general in that at can be extended to account for rotating stars as well, not that we only consider the vacuum surrounding a rotating black hole solution. A quicker but less conceptual derivation can be done 'analytically rotating' into complex coordinates covered in —[REF]—. One can also 'brute force' the calculation by considering an ellipsoidal ansatz [NP13], which is more intuitive but perhaps tedious in terms of computation. All of the above approaches result in the same metric regardless, given by the following expression:

$$ds^2 = -\frac{\Delta}{\Sigma}(dt - a\sin^2(\theta)d\phi)^2 + \frac{\sin^2(\theta)}{\Sigma^2}((r^2+a^2)d\phi - adt)^2 + \frac{\Sigma}{\Delta}dr^2 + \Sigma^2 d\theta^2 \tag{3.1}$$

Here, $\Delta = r^2 - 2Mr + a^2$, $\Sigma = r^2 + a^2\cos^2(\theta)$ and $a = \frac{J}{M}$ is the Kerr parameter.

3.2 Analysing the Geometry of Kerr Spacetime

3.3 Integrability of Kerr Geodesic Flow

Kerr geodesic flow has three obvious constants of motion. Since we have geodesic flow there is the Hamiltonian itself, or the rest mass of the particle. These arise from three obvious symmetries of the metric. Another conserved quantity arises from the axial symmetry generated by ∂_φ, implying that there is a conserved momentum $p_\varphi = g_{\mu\varphi}p^\mu$. Furthermore, there is a timelike killing field which induces a conserved p_t associated to the energy. There is in fact a third 'hidden symmetry'

associated to the total phase space $T^\star M$. We say hidden since this is not a lift of a symmetry in coniguration space, unlike usual translations along coordinates for example. The existence of such a transformation is usually hard to deduce, especially in 8-dimensional curved phase space.

Instead, we can go the other way and infer the existence of a global conserved quantity. This is known as Carter's constant. Then, by Noether's theorem, this does correspond to a symmetry vector field in phase space. This won't be as important to the following discussion however. Recall that the existence of four Poisson commuting and independant conserved quantities makes geodesic flow **integrable**. In particular, this means that we can now solve the geodesic equation exactly.

3.3.1 Hamilton-Jacobi equation

Chapter 4

Causality & Topology

4.1 Preliminaries

The main goal of this chapter will be to understand the causal nature of spacetime and cover singularity theorems. We will mainly follow Witten's lectures [Wit20] as well as Galloway's [Gal14]. Note that we won't be proving every statement carefully, such proofs can be found in [O'N83] and references therein. Let's begin with some definitions. A Lorentzian metric on a spacetime defines light cones, $\mathcal{V}_p$, in the tangent spaces at points by:

$$\mathcal{V}_p \coloneqq \{X \in T_pM : g_p(X, X) = 0\}$$

At each point there are actually two components of $\mathcal{V}_p$, and we need to distinguish one of these as future pointing. This assignment needs to smooth and defined over the whole spacetime to have a reasonable sense of the 'future'. Such an assignment is a time orientation.

Definition 13. *A Lorentzian manifold M is time orientable iff it admits a smooth timelike vector field $T \in \mathfrak{X}(M)$.*

Note that not all Lorentzian manifolds are time orientable, even if the underlying manifold is orientable. —-[GIVE EXAMPLE]—-. For curves in spacetime we can characterise them by three groups :

Definition 14. *A curve $\gamma : \mathbb{R} \longrightarrow M$ is*

1. ***Timelike*** *if $g(\dot{\gamma}, \dot{\gamma}) < 0$*
2. ***Null*** *if $g(\dot{\gamma}, \dot{\gamma}) = 0$*
3. ***Spacelike*** *if $g(\dot{\gamma}, \dot{\gamma}) > 0$*

   for all points in the image of γ.

We say that a curve is causal if it is either timelike or null. Intuitively, matter particles must travel along timelike curves and photons along null ones. This is just the statement that no particle can move faster than the speed of light locally. We can define a relation on events in spacetime based on whether they lie in each other futures or pasts :

Definition 15 (Causal relations)**.**

1. $p \ll q \iff \exists$ *a future directed timelike curve from* p *to* q
2. $p < q \iff \exists$ *a non-trivial future directed causal curve from* p *to* q

We then say that q is in the timelike/causal future of p. Denote $p \leq q$ if q is either in the causal future of p or $p = q$. We can define the regions of causality as follows :

Definition 16 (Causal regions)**.**

1. *The* ***timelike future*** *of a point* $p \in M$ *is* $I^+(p) \coloneqq \{q \in M : p << q\}$
2. *The* ***causal future*** *of a point* $p \in M$ *is* $J^+(p) \coloneqq \{q \in M : p \leq q\}$

We defined the case for the future regions, it can be easily extended to past regions by reversing time. For example $J^-(p)$ would be set of all q such that $q \leq p$. For the rest of the discussion we will construct definitions for future (+) events, but let it be implicit that there are also time dual past (-) definitions. Note that $I^+(p)$ is open $\forall p \in M$, but $J^+(p)$ may or may not be closed. —[EXAMPLE]—.

Proposition 1. *Suppose* $q \in J^+(p) - I^+(p)$*, then any future directed causal curve from* p *to* q *must be a null geodesic.*

From this proposition we have the following two statements :

Remark 3. *The causal relations are transitive, and we have the following:*

$$(p \ll q \ \wedge \ q < r) \ \vee \ (p < q \ \wedge \ q \ll r) \implies p \ll r$$

We can generalise the causal/timelike futures of points to arbitrary subspaces in spacetime.

Definition 17. *Consider a subset* $S \subseteq M$*.*

1. *The* ***timelike future*** *of* S *is*

$$I^+(S) \coloneqq \bigcup_{p \in S} I^+(p) = \{q \in M : p \ll q \quad \text{for some} \quad s \in S\}$$

2. *The* ***causal future*** *of* S *is*

$$J^+(S) \coloneqq \bigcup_{p \in M} J^+(p) = \{q \in M : p \leq q \quad \text{for some} \quad s \in S\}$$

Definition 18. *A submanifold of spacetime,* $\iota : \Sigma \hookrightarrow M$*, is a spacelike hypersurface if* $\iota^\star g$ *has Riemannian signature.*

—WITTEN PAGE 12— —TALK ABOUT CYLINDER $S^1 \times \mathbb{R}_t$ EXAMPLE WITH A LINE WRAPPING AROUND.—

Definition 19. *A subset* Σ *of spacetime is* ***achronal*** *if there exists no timelike curve connecting two arbitrary points* $p, q \in \Sigma$

Definition 20. *An* ***achronal boundary*** *is a subset of the form* $\partial I^{\pm}(S)$ *for some achronal subspace* S*.*

Proposition 2. *If an achronal boundary is non-empty,* $\partial I^{\pm}(S) \neq \varnothing$*, then is it a closed achronal* C^0 *hypersurface in* M

Note that by C^0 hypersurface we mean a topological submanifold of codimension one. Achronal boundaries need not be smooth in general.

Definition 21. *Suppose* $S \subseteq M$ *is achronal.* $p \in \bar{S}$ *is an* ***edge point*** *of* S *if every neighbourhood* $U \ni p$ *contains a timelike curve,* γ*, from* $I^-(p) \cap U$ *to* $I^+(p) \cap U$ *that does not meet* S*,* $\gamma \cap S = \varnothing$*. Denote the set of edge points* $\mathrm{edge}(S)$*. We say that* S *is* ***edgeless*** *if* $\mathrm{edge}(S) = \varnothing$*.*

—-[GIVE TWO EXAMPLES]—.

Proposition 3. *An edgeless achronal set,* $S : \mathrm{edge}(S) = \varnothing$*, is a* C^0 *hypersurface.*

—-PROOF IN O'NIEL—-

4.2 Causality Conditions

Definition 22. *The* ***chronology condition*** *states there are no closed* ***timelike*** *curves in a spacetime* M*.*

Definition 23. *The* ***strong causality condition*** *states that there a non-trivial closed* ***causal*** *curves in a spacetime* M*.*

4.3 Hawking's Theorem

4.4 Penrose's Theorem

4.5 String Theory and Instability

Bibliography

[Gal14] G. J. Galloway. Notes on lorentzian causality. ESI–EMS–IAMP Summer School on Mathematical Relativity, Vienna, Austria, August 2014. Unpublished lecture notes.

[NP13] Branislav D. Nikolić and Milan R. Pantić. A possible intuitive derivation of the kerr metric in orthogonal form based on ellipsoidal metric ansatz, 2013.

[O'N83] Barrett O'Neill. *Semi-Riemannian Geometry: With Applications to Relativity*, volume 103 of *Pure and Applied Mathematics*. Academic Press, New York, 1983.

[Str13] Norbert Straumann. *General Relativity*. Springer, Dordrecht, 2013.

[Wit20] Edward Witten. Light rays, singularities, and all that. *Rev. Mod. Phys.*, 92:045004, Nov 2020.